

Student Answer Sheet

Observability & Centralized Logging
(Chapter 5)

Course: Systems Reliability Engineering (SRE)

Lab: Observability & Centralized Logging

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5

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1 The Challenge (Section 2)

Explain why binary “up/down” monitoring failed in this instance. Why is monitoring “pre-defined questions” insufficient for identifying p99 latency spikes in a distributed microservices environment?

Your Analysis:

2 Part 1: The Three Pillars (Section 3)

2.1 Pillar Characteristics Table

Complete the table below:

Three Pillars of Observability		
Pillar Name	Data Characteristic (Discrete vs. Aggregated, Cardinality)	Specific Use Case (E-Commerce Scenario)
Logs	_____	_____
Metrics	_____	_____
Traces	_____	_____

2.2 Diagnostic Question (Section 3.1)

How do **Traces** specifically resolve the problem of identifying bottlenecks in a distributed request lifecycle where every individual service appears healthy but the overall response is delayed?

Your Answer:

3 Part 2: ELK Technical Analysis (Section 4)

3.1 Beats & Elastic Agent

Your Answer:

3.2 Logstash & Grok

Your Answer:

3.3 Elasticsearch & The Inverted Index

Your Answer:

3.4 Kibana

Your Answer:

4 Part 3: WEF Design Challenge (Section 5)

4.1 Architecture Evaluation Table

WEF Architecture Comparison		
Feature	Source-Initiated (Push)	Collector-Initiated (Pull)
Scalability	_____	_____
Firewall Configuration	_____	_____
Suitability for Remote Laptops	_____	_____

4.2 Windows Server 2025 Innovation (Section 5.2.1)

Your Answer:

4.3 Forensic Prioritization (Section 5.2.2)

Identify the two specific Windows Event IDs for logon success and failure:

Logon Success Event ID: _____

Logon Failure Event ID: _____

5 Part 4: SLIs and Error Budgets (Section 6)

5.1 Error Budget Calculation (Section 6.1)

Given: 99.95% SLO. Total minutes in 30 days = 43,200.

Show your work:

Error Budget = _____

Final Answer: _____ minutes per month.

5.2 SLI Proposal (Section 6.2)

Your Answer:

5.3 The “SRE Clamp” (Section 6.3)

Your Answer:

6 Part 5: Practical Implementation (Section 7)

6.1 Task 5.1: docker-compose.yml

Listing 1: docker-compose.yml

```
1 # TODO: Write your Docker Compose configuration here
2 version: '3.8'
3 services:
4   # elasticsearch:
5   #   ...
6   # logstash:
7   #   ...
8   # kibana:
9   #   ...
```


6.2 Task 5.2: filebeat.yml

Listing 2: filebeat.yml

```
1 # TODO: Write your Filebeat configuration here
2 # filebeat.inputs:
3 #   ...
4 # output.logstash:
5 #   ...
```

6.3 Task 5.3: winlogbeat.yml

Listing 3: winlogbeat.yml

```
1 # TODO: Write your Winlogbeat configuration here
2 # winlogbeat.event_logs:
3 #   ...
4 # output.logstash:
5 #   ...
```

6.4 Task 5.4: logstash.conf

Listing 4: logstash.conf

```
1 # TODO: Write your Logstash pipeline configuration here
2 # input { beats { port => 5044 } }
3 # filter { grok { ... } }
4 # output { elasticsearch { ... } }
```

6.5 Task 5.5: Visual Proof

[Attach Kibana dashboard screenshot here]

7 Final Deliverable: System Interaction Map (Section 8)

Complete one template per data source (minimum three).

Data Source 1

Source Name: _____

Telemetry Type: (Log / Metric / Trace) _____

Shipper: _____

Identity Protocol/Tool: _____

Processing: _____

Transformation Step: _____

Final Destination: _____

Storage/Visualization Tool: _____

Data Source 2

Source Name: _____

Telemetry Type: (Log / Metric / Trace) _____

Shipper: _____

Identity Protocol/Tool: _____

Processing: _____

Transformation Step: _____

Final Destination: _____

Storage/Visualization Tool: _____

Data Source 3**Source Name:** _____**Telemetry Type:** (Log / Metric / Trace) _____**Shipper:** _____**Identity Protocol/Tool:** _____**Processing:** _____**Transformation Step:** _____**Final Destination:** _____**Storage/Visualization Tool:** _____